13. URBAN BASIC SERVICES

This sub-section reviews the present status with regards to provision of urban basic services in Greater Mumbai area

13.1. Water Supply

13.1.1 Existing Situation

The Hydraulic Engineering Department of MCGM is responsible for water supply in Greater Mumbai.

The population of Mumbai in the year has reached nearly 12 million with water supply requirement of around 3900 Million Liters per Day (MLD). At present the domestic, commercial, and industrial supply is catered to the tune of 3,100 MLD. Lakes are created by impounding rainwater by constructing dams across the rivers and valleys.

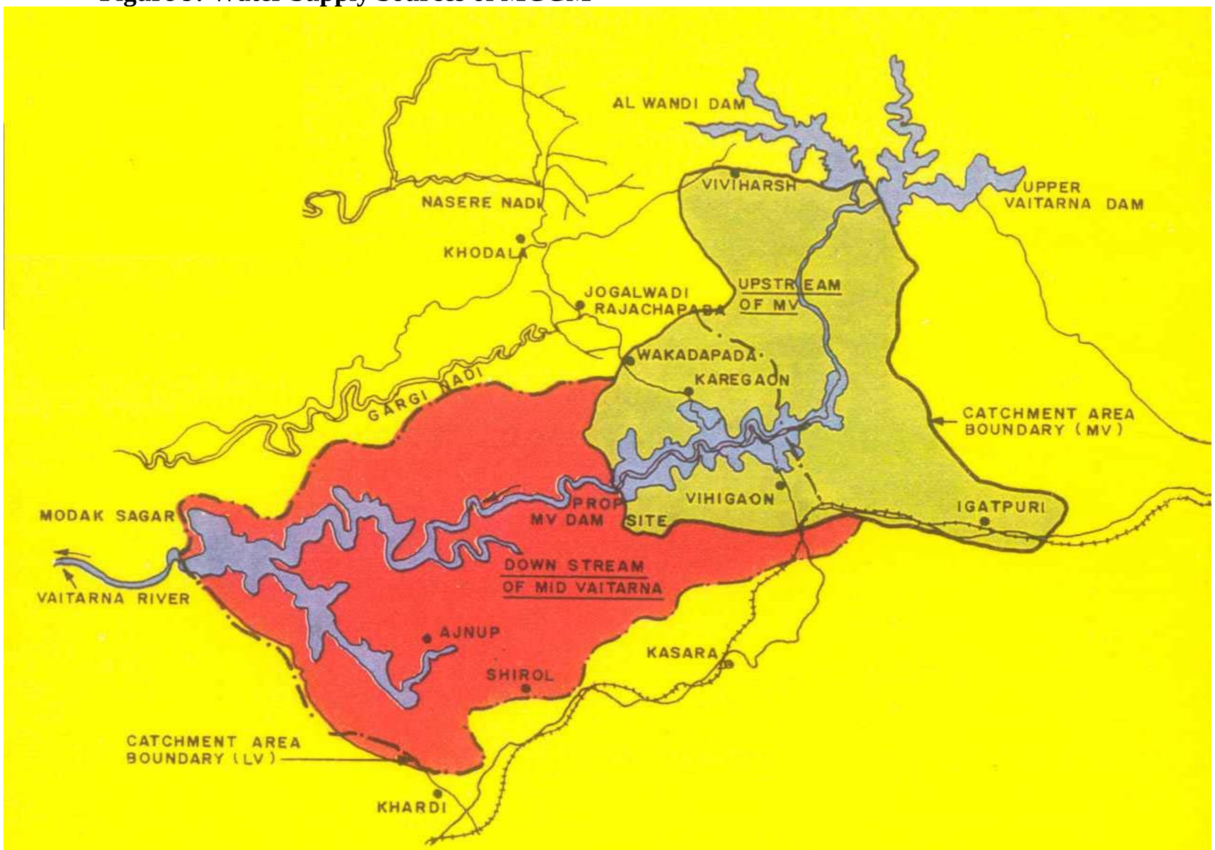
Table 34: Water Sources of MCGM

Sl.	Source	Yield in (MLD)	Ownership	Distance from City	Treatment Plant Location
1	Tulsi	18	MCGM	Within City Limit	Tulsi
2	Vihar	110	MCGM	Within City Limit	Vihar
3	Tansa	417	MCGM	100 Km from City	Bhandup Complex
4	Upper Vaitarna	1,025	MCGM	110Km from City	Bhandup Complex
5	Bhatsa	1,650	GoM	100 Km from City	Bhandup Complex/ Panjarapur
6	***Sub-total***	***3,220***			
7	En-route supply	-120			
	Total supply to city	**3,100**			

The water is completely treated with pre-chlorination, alum-dosing, settling, filtration and post-chlorination before supplying to consumers. The back wash quantity generated at Bhandup treatment works (which is about 45 MLD) is released in the Vihar Lake. However the backwash quantity generated at Panjrapur treatment works (approx. 20 MLD) is discharged in nallahs.

The treated water is stored in the Master Balancing Reservoirs (MBRI) at Bhandup Complex (246 ML) and MBRI at Yewai (123 ML) and further distributed to 28 service reservoirs spread through the city by a complex network of inlet mains which are maintained charged for 24 hours. These service reservoirs in turn supply water to the consumers in different water supply zones at suitable time for the duration varying from 24-hours to 90 minutes depending on area of the zone, topography, etc. The pressure in distribution system is in the range of 1 to 1.5 kg/cm^2 during water supply hours. The transmission system comprises about 300 km of primary transmission pipeline and about 650 km of secondary transmission pipeline.

Figure 9: Water Supply Sources of MCGM



A. Water Supply Sources / Schemes for Mumbai

a. *Vihar Scheme*:

Vihar scheme was the first piped water supply scheme and was commissioned in 1860. Three earthen dams and a masonry overflow section was constructed to impound Mithi river water about 20 Km north of Mumbai city. The impoundage is known as Vihar Lake and is a popular picnic spot. A 1200 mm dia cost iron pipe was laid to supply 32 MLD to a population of 700 thousands. Supply from this source was increased to 68 MLD by raising of dam in 1872 this scheme is still in operation.

b. *Tulsi & Powai Scheme*

In 1885 it was decided to develop Tansa as a next source of water supply, however due to critical situation of water shortage led to immediate development of Tulsi lake, up stream of Vihar lake on the Mithi river. This scheme consisted of an earthen dam, a masonry dam and 600 mm dia pipeline to carry 18 MLD water to the service reservoir at Malbar Hill in South Mumbai. While Tansa scheme was under construction Powai scheme was taken up on emergency basis on a tributary of Mithi River due to anticipated water famine in 1891. This scheme supplies 4 LD water, however due to inferior quality of water it is used for dairy and agricultural use only.

c. Tansa Scheme

Today, The four major source of water supply to the Mumbai City are Tansa, Vaitarna (Modak Sagar), Upper Vaitarna and Bhatsa. Tansa and Vaitarna are fully owned and constructed by MCBM only for City's water supply, whereas Upper Vaitarna and Bhatsa are multipurpose schemes and constructed and owned by Govt. of Maharashtra. Tansa lake is situated near Shahapur in Thane District about 100 Km away from Mumbai City. The construction of Tansa dam was started in 1886 and the scheme was fully developed in four stages. The first stage was completed in 1892. A dam of 2800 m in length and having maximum height about 40 m was constructed in the first stage. Also a 42 Km long closed duct line on the hills and 53 Km long 1200 mm dia cast iron pipeline through valleys was laid to convey 77 MLD water to the heart of City.

The second stage was commissioned in 1915 by an additional pipeline of 1250 mm dia and supply was increased by 82 MLD. In the third stage height of dam was increased by 3 m to increase the storage and new conveyance system of two 1800 mm dia M.S. pipe lines were laid along a shorter route and was commissioned in 1925. The supply was increased by 68 MLD. The fourth and last stage was completed in 1948. In this stage storage capacity was increased by providing 38 nos. of floodgates of 50' x 4' size. The storage capacity of the dam is 40,666 million gallons at Full supply level i.e. 422' THD. The water is drawn through the seven outlets at different levels i.e. at 415', 410', 405', 400', 390', 380', & 375' THD. The total water supply from this source is about 410 MLD.

d. Lower Vaitarna Scheme

After the commissioning of Tansa Scheme in 1948 Mumbai's water supply was about 495 MLD, but after independence there was a large influx of the people in the city and also there was unprecedented increase in industrial and commercial activity due to which demand of water supply increased considerably. Therefore it was necessary to augment the water supply to the City and hence it was decided to tap the Vaitarna river water by the Municipal administration at that stage. A scheme known as Vaitarna cum Tansa was conceived, planned and executed by a team of municipal engineers under the able guidance and leadership of late Shri. N.V.Modak, then City Engineer of MCGM. In this scheme a 90 m high and 500 m long dam was constructed across river Vaitarna. The dam was the first concrete dam in the country, which was constructed using pre-cooled concrete. The impoundage created by this dam was subsequently named as Modak Sagar in the recognition of valuable services of late Shri. N.V.Modak. The capacity of the reservoir is 46000 MG. The catchment area of dam is 174 sq miles and the water-spread area is 3.8 sq.miles with a length of lake about 10 miles. The water is released through intake tower, which is typical in design and construction and provided with controlled opening through which a calculated amount of water can be released in the pipe. The openings are provided at different elevations from the bottom i.e. at 416' THD, 1 No. 460' THD 2 Nos., 480' THD 2 Nos., 500'THD 2

Nos., and 520' THD 2 Nos.

This arrangement of openings allows water to be drawn from the level which at any time will supply water having the least turbidity, algae or least amount odor and bad taste. To bring the water from lake, a 7.2 Km. tunnel was constructed between Vaitarna and Tansa and a M.S. pipeline partly 3000 mm. and partly 2400 mm. dia. 87 Km. long was laid from Tansa to City. A new concept in pipeline design using ring girders and roller supports with expansion joints was used for above ground pipelines. This brought considerable saving in the M.S. plate at that time. This pipeline was the largest and longest pipeline ever laid in the world. Tansa Lake from the lower gate provided in the bifurcation chamber which is constructed at the end of the tunnel. The surplus water from Modak Sagar can be thus diverted to fill up Tansa Lake, which does not fill up in dry years.

This scheme was completed in 1957 augmenting City's water supply by additional 540 MLD.

e. Upper Vaitarna Scheme

Govt. of Maharashtra (GOM) in the year 1960 took up the execution of a multipurpose scheme known as Upper Vaitarna scheme. Under this scheme an impoudage was created in the upper reaches of Vaitarna River by constructing two dams on river Alwandi & Vaitarna about 80 Km. u/s. of Modak Sagar. The impounded water is used to generate electricity through hydroelectric power station of 60 MW capacity and then it is released in the Vaitarna River to flow down to the lower Vaitarna Lake. (Modak Sagar). This released water is used for water supply by conveying it to the City. The GOM constructed the dam while M.C.B.M. carried out the works of laying of partly 3000 mm dia. and partly 2750 mm. dia. trunk main and construction of some service reservoirs. The full supply level of upper Vaitarna reservoir is 1980' MSL i.e.2060.25' THD and the capacity is 73,125 Million gallons. Under this scheme a tunnel under Thane Creek was also bored to carry the entire water supply under creek from security point of view. This project was fully commissioned in the year 1972 and the water supply was increased by 540 MLD. Upto this point it was a unique feature of Mumbai's water supply system that it was conveyed entirely by gravity. The catchment area of Tansa, Vaitarna and Upper Vaitarna were so protected that the water hardly need any filtration and only chlorination was done for disinfections purpose.

f. Bhatsa Scheme

Despite the implementation of the aforementioned schemes, large-scale shortages in water supply started arising towards the end of 1960s. The rate of growth of the population during the decades of 1950s and 1960s was unprecedented. Therefore a Master plan was prepared for integrated development of water supply and sewerage facilities. This project was titled Mumbai Water Supply and Sewerage project. M/s. Binnie & Partner of London was appointed as consultant for this project. This project was implemented in three stages, with the financial assistance from IDA and World Bank. In each of these three stages 455 MLD water was drawn from Bhatsa River. Government of Maharashtra constructed a dam across river Bhatsa near Shahpur in Thane district and works of pumping, conveyance, treatment, storage were carried out by M.C.B.M.

g. I Mumbai Water Supply Project (Bhatsa Stage I)

Various alternative methods for developing the Bhatsa Scheme were examined. In each of the alternative it was common that the Bhatsa reservoir would be used as regulating and storage reservoir and the water would be abstracted from the river at short distance upstream of Kalu confluence and the water would be treated at Bhandup. The alternatives were differed only in type of conveyance system to be adopted. The different degrees of integration with the Vaitarna and upper Vaitarna mains, completely independent operation or completely independent operation except for Bassin creek. As a result of economic studies schemes was adopted in which water from Bhatsa was to be injected in to Vaitarna and Upper Vaitarna main at a point close to the point of abstraction at Pise. In Bhatsa stage I, following project components were constructed.

- A pick up weir across the Bhatsa River at Pise, just upstream of its confluence with the Kalu and near the upper limit of tidal effects.
- An intake channel leading to an intake pumping station on the right bank of the river housing 7 vertical spindle mixed flow pumps (5 duty and 2 standby) each with a nominal output of 91 MLD against a total lift of 97.2 m (319 ft.).
- A 2235 mm. (88 in) welded steel rising main 9.2 KM (5.7 miles) long, laid underground except for one bridge crossing, between the pumping station and the Vaitarna/Upper Vaitarna mains at Yewai.
- A treatment plant comprising of settling tanks and a pumping station comprising of 6 Nos. 91 MLD + 2 timmer 45 MLD pump sets at Panjrapur.
- A control valve complex at Agra Road Yard, a short distance upstream of Yewai on the trunk mains alignment, where the gravity flow in the trunk mains is regulated.
- A new tunnel, 5 m (16.4 ft.) dia. to carry water from the trunk mains to the Bhandup treatment works, and new mains to reinforce the existing East-West tunnel system.
- A treatment works sited on land on the shore of Vihar Lake, for all the water from the mainland sources; consisting of 5 blocks of 24 filters operating on the declining rate principle, together with associated chemical storage and dosing facilities, a chlorine contact tank, access roads, quarters, etc
- A treated water pumping station containing 10 (7 duty + 3 standby) 245 MLD (vertical spindle, double suction, split-casing pumps, delivering to a new master balancing reservoir of 245 ML capacity and 7 (5 duty + 2 standby) 38 MLD similar pumps, delivering to Malad and Borivli reservoirs, through the second leg of the East-West tunnel; and

- Treated water mains, 2 x 2750 mm (108 in) to the master balancing reservoir and 1 x 1. 350 (54 in) main to the East-West tunnel.

The first stage was commissioned in 1981 augmenting city's water supply by 455 and total water supply to the City after this stage was 1970 MLD.

h. *II Mumbai Water Supply Project (Bhatsa Stage II)*

In stage – I Water from Bhatsa was directly injected into existing Vaitarna & upper Vaitarna mains after sedimentation. However in stage II water from Bhatsa river is fully treated at Panjrapur and stored in MBR – II at Yewai hills and transmitted to the distribution system in City at MGL & E.E.H. junction. In Bhatsa stage II following works were undertaken.

- Expansion of pumping station at Pise and installation of 7 pumps (4 working + 3 stand by) of capacity 113.6 MLD each.
- Construction of pre chlorination house at Pise
- Laying of an additional 2235 mm dia M.S. above ground pipeline from Pise to Panjrapur.
- Construction of full fledge 455 MLD capacity treatment plant at Panjrapur and pumping station having 7 pump sets each having 113.6 MLD capacity.
- Construction of a new master-balancing reservoir (MBR II) having 117 M.L. capacity.
- Laying of 48 KM long 2235 mm dia transmission main from Yewai MBR upto City.
- Construction of 3500 mm dia. tunnel under Kasheli Creek.
- Construction of 3000 mm dia. tunnel from Mahalaxmi racecourse to Malabar Hill using tunnel-boring machine. The tunnel-boring machine for tunneling was used for the first time in India.
- Construction of service reservoir at Pali Hill Ghatkopar Hills and mortar lining of new as well as old mains was also undertaken to improve the 'C' value of pipe surface thereby increasing the discharging capacity of the mains.

This scheme was commissioned in 1989 and augment City's water supply by another 455 MLD to 2425 MLD.

i. *III Mumbai Water Supply Project*

In this stage following works were undertaken.

- Expansion of pumping station and installation of 7 pump sets each of 113.6 MLD capacity.

- Laying of an additional 3000 mm dia. above ground pipeline from Pise to Panjrapur.
- Full two stage 455 MLD capacity treatment plant and pumping plant at Panjrapur.
- Laying of 3000 mm dia. transmission main from Yewai MBR – II to City.
- Construction of Reservoir at Bhandup (BUDP), Borivali Hills, Malabar Hills, Worli Hills and Raoli Hills.

This scheme was commissioned in 1997 augmenting City’s water supply to about 2900 MLD.

j. <u>III A Mumbai Water Supply Project</u>

Due to delayed monsoon in 1992, the Govt. of Maharashtra (GoM.) appointed an Expert Committee under the Chairmanship of Dr. M.A.Chitale, to suggest immediate measures, and also the long term strategy to meet the water supply needs of Greater Mumbai. The Committee’s report has been accepted by the GoM, and all the sources recommended in the report have been allocated to MCGM by the Irrigation Department (I.D.) GoM. The Committee has projected the population of Mumbai for the year 2021 as 1.56 Crore and suggested following water sources to be developed priority wise for satisfying the water demand.

- Unutilized Irrigation potential of Bhatsa (III-A Mumbai Water Supply Project)
- Middle Vaitarna Project. (IV Mumbai Water Supply Project.)
- Kalu Project
- Gaargai Project
- Shahi Project
- Pinjal Project

Accordingly, III-A Mumbai Water Supply Scheme (i.e. augmenting 455 MLD water by abstracting unutilized irrigation potential of Bhatsa Dam) is already launched. Under III A Mumbai Water Supply Project additional 455 MLD water will be abstracted at Pise, pumped to Panjrapur and from thereon injected into existing Vaitarna mains for conveying towards Bhandup for treatment. At the same time raw water of Mumbai III Project will be treated in a new treatment plant at Panjrapur. In the meantime GOM has allowed Thane Municipal Corporation to abstract 100 MLD of water out of 455 MLD allotted to BMC for short period. However, the project components are designed for abstraction, conveyance and treatment of 455 MLD of water.

The following project components are undertaken under III A Mumbai Water Supply

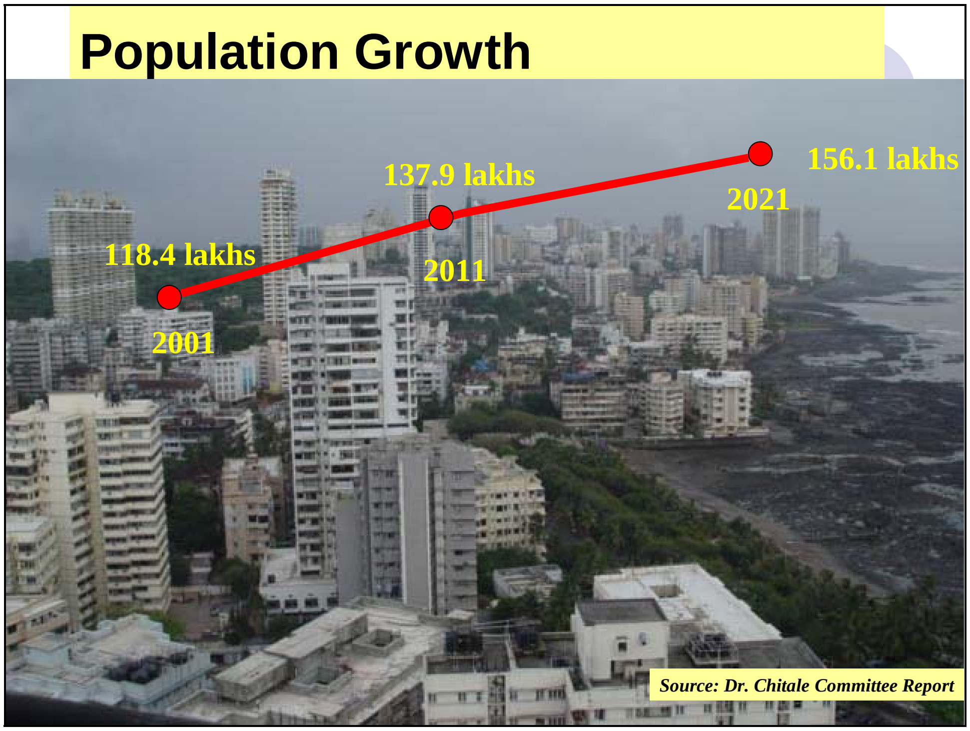
Project.

- Raising of crest level of existing Pise weir by 2.0 b using inflable gates to increase capacity of pond.
- Pumping stations of capacity 455 MLD each at Pise and Panjrapur.
- Rising main of 3000 mm dia. 9.5 KM length from Pise to Panjrapur.
- Construction of water treatment plant comprising of filters, clarifiers etc. in the existing Panjrapur complex to treat Mumbai III water.
- Supply, installation and commissioning of pumping plants including associated electrical works at Pise and Panjrapur.
- Treated water transmission main of 2400mm. dia from Panjrapur Stage III Pumping station to Yewai MBR.
- Transmission main of 3000 mm. dia. from MGL/EEH Junction to Vikhroli and further up to Amarmahal for total length of 13 Km.
- A 3000 mm dia. tunnel from Bhandup to Malad reservoir, Malad reservoir to Liberty Garden and further up to Charkop having total length of 12 KM.
- A 2400 mm dia tunnel from Veravali Reservoir to Adarsh Nagar and further upto Yari Road:

The total cost of the project is Rs. 943 crore. The project is likely to be completed by 2006 and additional 455 MLD water will be brought to the City. Out of this about 150-200 MLD is already being brought to City.

B. Population Projections and Water Demand

The populations figures for 2001 based on the demographic survey conducted by the Govt. of India was 11.84 million and it is expected to go up to 13.79 million in 2011 and to 15.61 million in 2021 as per Chitale committee. The projections have been made by the incremental increase method.



Population projections up to year 2011 indicate a demand of 5043 Mld and 5388 Mld in the year 2021. In arriving at these figures the average per capita demand adopted by Chitale committee is 240 lpcd.

The gap between demand and supply was 1642 Mld in 2001 and the same is expected to be 1210 Mld in 2011 and to 1100 Mld in 2021.

C. Water Supply and Demand Projections

Year	Increment	Supply (Mld)	Demand (Mld)	(%) fulfillment	Remarks
2001		3025	3975	76	
2005	150	3175	4150	76	Partial completion of IIIA
2007	200	3375	4300	78	Completion of IIIA
2011	477	3852	4526	85	Completion of Mid Vaiterna
2016	455	4307	4800	90	Gargai
2021	765	5172	5068	100	Pinjal

D. Distribution Network

The distribution network has been laid and upgraded over the past 136 years or so. The distribution system includes the 28 service reservoirs and allied piping systems for inlets and outlets feeding 110 water supply zones. Water mains amounting to approximately 4,000 km in length, range in diameter from 80 mm to 1,800 mm. Water mains have are either cast iron (CI), mild steel (MS) or ductile iron (DI), some of which have internal lining. The MS pipes are externally protected by either concrete or other proprietary coatings.

There are a total 2,94,200 metered connections of which 2,42,000 (83%) are domestic connections including 1,62,000 connections in slums, 47,000 (15%) commercial connections and 5,200 (2%) industrial connections. There are about 75,000 un-metered connections, primarily in the core areas of the island city area. Of the total metered consumption of water, about 87 per cent is consumed by domestic consumers, 9 per cent by commercial consumers and about 4 per cent by commercial consumers.

Water quality is maintained through regular collection and testing of water samples throughout the network and from customer taps.

Due to the fact that demand exceeds supply, an intermittent supply regime is operated whereby each supply zone is ‘charged’ with treated water for between 2 and 6 hours per 24-hour period. This is a root cause of contamination due to ingress of foul water during non-supply hours through joints, disused connections, tampered mains, faulty fittings, etc. Though adequate care is taken to monitor the water quality at all the points from source to end point at consumer’s tap, incidents of contamination do occur. Fortunately all the sources have fairly protected catchments and no chemical contamination was observed at the source.

The water supply system of Mumbai is now equipped with Supervisory Control and Data Acquisition (SCADA). The water billing & revenue collection system is also computerized in all the wards. Supplies into each zone are controlled and monitored by the three local control centers (LCCs) located in the City, Western Suburbs and Eastern Suburbs

Calculation of ‘water balance’ is difficult due to insufficient ‘zonal’ and ‘district’ meters, as well as the lack of metered outlets at reservoirs. It is estimated that about 80 % of consumer meters are not operable or reliable.

In spite of numerous constraints and limitations the corporation is successful in supplying regular and adequate quantity to the citizens of Mumbai. There are occasions of pipe bursts or power failure, when water supply is interrupted. But in all cases, water supply is restored in shortest possible time, which rarely exceeds 12-hours in case of major burst. This year 272 Nos. of leakages & 60 Nos. of bursts were detected & repaired on water mains of 600 mm diameter & above. Also this year 28,720 Nos. of visible leakages were detected &

28,684 Nos. of leakages were repaired.

The water supply to the city totally depends on rainfall during the monsoon and as such water cut of 10% to 15% required to be imposed in case of bad monsoon or delayed monsoon. Some of the industries and commercial establishments (e.g. RCF & Railways) recycle used water and reuse the same for non-potable purpose.

E. Proposed Water Supply Schemes

IV- Mumbai Middle Vaitarna Water Supply Project

As per the recommendation of the Expert Committee, the Middle Vaitarna Project was envisaged and a detailed feasibility study was carried out for the Middle Vaitarna Dam (in 1992) and the Conveyance and Treatment components of the scheme. The project involves creation of an impoundage by constructing a Dam on Vaitarna River between Upper and lower Vaitarna dams. This project, will augment water supply to Mumbai by further 455 MLD thereby making the total water supply to 3810 MLD. The project has been sanctioned by Irrigation Department of Govt. of Maharashtra in 1997.

Salient features of the Middle Vaitarna Project are:

The Proposed Middle Vaitarna Dam (M.V.D.) on the Vaitarna River is having an approximate length of 500 m and height of about 100 m.

This will be a R.C.C. (Roller Compacted Concrete) Dam.

The water from the MVD will be conveyed through the river upto Modak Sagar Reservoir i.e. upto Lower Vaitarna reservoir.

From Modak Sagar, Water will be drawn through a new intake tower in a controlled manner and will be a conveyed through 7.5 K.M. long tunnel upto Intake Shaft at Belh Nallah.

Thereafter through 3000 mm dia. M.S. Pipeline the water will be brought to the Bhandup Complex.

This additional water will be treated at Bhandup Complex by constructing a New Water Treatment Plant having capacity of 500 MLD. This filtered water will be pumped by proposed pumping station to proposed M.B.R. at Bhandup Complex.

From this M.B.R., water will be distributed to Mumbai.

The Project is expected to be completed within 5 to 6 years from inception, with a total cost of about Rs. 1600 Crore.

13.1.2 Key Issues and Strategy Options/Plans

Key Issues	Strategy Options / Plans
<u>Inadequate Source</u> The current supply of 3,050 MLD is inadequate to meet the current demand of about 3,900 MLD. Demand for water is projected to increase to about 4,526 MLD by 2011 and to 5,068 MLD by 2021	Implement recommendations regarding water allocation, of the Expert Committee Report under the Chairmanship of Dr. MA Chitale, 1993. The sources recommended to be developed for supply to Mumbai have been allocated by the Irrigation Department, GoM The report has recommended the following sources, which would be adequate to meet the estimated gap of 2000 MLD by 2021 Unutilized Irrigation potential of Bhatsa (III-A Mumbai Water Supply Project) – 455 MLD by 2007 – project implementation is ongoing Middle Vaitarna Project. (IV Mumbai Water Supply Project.) – 477 MLD by 2011 – DPR prepared in year 2002 – to be taken up for implementation immediately. Estimated project cost is Rs. 1,351 crore Gaargai Project – 455 MLD to be implemented by 2016 Pinjal Project – 765 MLD to be implemented by 2021
Bad condition of Transmission & Distribution System– leading to leakage and contamination of water	Implement recommendations presented in the study by M/s Binnie Black & Veatch, Thames Water and TCE Consulting Engineers for primary transmission. Implement the five-year priority works plan for improving the distribution system. Based on in-house experience and knowledge, MCGM has prepared a five-year plan for priority works to be undertaken to improve the distribution system performance. This would essentially comprise rehabilitation and replacement of pipes. Rs. 500 crore have been sanctioned for this purpose with annual investment of Rs. 100 crore per year to replace/rehabilitate old water mains of be taken up from FY 2006-07.
Absence of water balance information – resulting in inequitable supply and unaccountable water in the system	Install zonal and district meters at appropriate locations to facilitate regular water balance and audit analyses, aimed at facilitating equitable water supply and leak detection